ODE Assignment-2

1.1. (a) Let $u(t) = \sum_{k=0}^{\infty} c_k t^k$ be the solution for the given

ODE : $u''(t) - tu'(t) + u(t) = 0$ ...(i)

Now, $u'(t) = \sum_{k=0}^{\infty} c_{k+1}(k+1)t^k \Rightarrow tu'(t) = \sum_{k=0}^{\infty} c_{k+1}(k+1)t^{k+1}$

$\therefore tu'(t) = \sum_{k=1}^{\infty} c_k k t^k$

and $u''(t) = \sum_{k=0}^{\infty} c_{k+2}(k+2)(k+1)t^k$.

Substituting back $tu'(t)$, $u''(t)$ and $u(t)$ in eq(i), we get

$$\Rightarrow \sum_{k=0}^{\infty} c_{k+2}(k+2)(k+1)t^k - \sum_{k=1}^{\infty} c_k \cdot k \cdot t^k + \sum_{k=0}^{\infty} c_k t^k = 0$$

To collect the coefficient of $t^0, t^1, t^2, \ldots, t^k$, we get

at $k=0$, $c_2(2)(1) + c_0 = 0 \Rightarrow c_2 = -\frac{1}{2}c_0$

at $k=1$, $c_3(3)(2) - \cancel{c_1} + \cancel{c_1} = 0 \Rightarrow c_3 = 0$

at $k=2$, $c_4 \times 12 - c_2 \cdot 2 + c_2 = 0 \Rightarrow c_2 = 12c_4 \Rightarrow c_4 = -\frac{1}{24}c_0$

⋮

coefficient of t^k.

$$c_{k+2}(k+2)(k+1) = c_k(k-1)$$

$$\Rightarrow c_{k+2} = \frac{(k-1)c_k}{(k+2)(k+1)}, \quad (k \geq 1) \quad \text{...(ii)}$$

which is a recurrence between c_{k+2} & c_k

So, the even solutions,

$u_1(t) = c_0 + c_2t^2 + c_4t^4 + c_6t^6 + \cdots$ (even solution)

$= c_0 - \frac{1}{2}c_0t^2 - \frac{1}{24}c_0t^4 + \cdots$

$= 1 - \frac{1}{2}t^2 - \frac{1}{24}t^4 + \cdots$ (where $c_0 = 1$ in particular)

For the odd-solutions,

$$u_2(t) = c_1 t + c_3 t^3 + c_5 t^5 + \cdots$$

using eq (ii), we get

at $k=3$, $c_5 = \frac{2}{(5).(4)} \cdot c_3 = 0$

at $k=5$, $c_7 = \frac{4}{(7)(6)} \cdot c_5 = 0$

$\vdots$

at $k=2m+1$ $\quad c_{2m+3} = \frac{2m}{(2m+3)(2m+2)} \cdot c_{2m+1} = 0$ $\left(\because c_{2m+1} = 0 \text{ for each } m \in \mathbb{N}\right)$

Take $c_1 = 1$ (in particular).

$\therefore u_2(t) = t$ is the odd solution.

Now, to check whether u_1 & u_2 are linearly independent,

at $t=0$, $u_1(0) = 1$; $u_1'(t) = -t - \frac{1}{6}t^3 + \cdots$

$\Rightarrow u_1'(0) = 0$

And, $u_2(0) = 0$; $u_2'(t) = 1 \Rightarrow u_2'(0) = 1$.

$$\therefore W(0) = \begin{vmatrix} u_1(0) & u_2(0) \\ u_1'(0) & u_2'(0) \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \neq 0$$

Hence, u_1, u_2 are linearly independent $\forall t \in \mathbb{R}$.

($\because$ Wronskian cannot anywhere, if it's non-zero at some point).

So, by ratio test for $k=n$, where $n \in \mathbb{Z}$

$$R = \lim_{n\to\infty} \left| \frac{c_n}{c_{n+2}} \right| = \lim_{n\to\infty} \left| \frac{(n+2)(n+1)}{(n-1)} \right| \to \infty$$

$\therefore u_1(t)$ is convergent $\forall t \in \mathbb{R}$

& $u_2(t)$ is convergent $\forall t \in \mathbb{R}$.

(b) Let $u(t) = \sum_{k=0}^{\infty} C_k t^k$ be the solution for the given ODE: $u''(t) + t^3 u'(t) + t^2 u(t) = 0$

$$\Rightarrow \quad u'(t) = \sum_{k=0}^{\infty} C_{k+1}(k+1)t^k \Rightarrow t^3 u'(t) = \sum_{k=0}^{\infty} C_{k+1}(k+1)t^{k+3}$$

$$= \sum_{k=3}^{\infty} C_{k-2}(k-2)t^k$$

And,

$$u''(t) = \sum_{k=0}^{\infty} C_{k+2}(k+2)(k+1)t^k \; ; t^2 u(t) = \sum_{k=2}^{\infty} C_{k-2} t^k$$

Substituting back to ODE, we get

$$= \sum_{k=0}^{\infty} C_{k+2}(k+2)(k+1)t^k + \sum_{k=3}^{\infty} C_{k-2}(k-2)t^k + \sum_{k=2}^{\infty} C_{k-2} t^k = 0$$

To collect the coefficient of t^0, t^1, t^2, t^3, we get

at $k=0$, $2C_2 = 0 \Rightarrow C_2 = 0$

at $k=1$, $6C_3 = 0 \Rightarrow C_3 = 0$

at $k=2$,

$12C_4 + C_0 = 0 \Rightarrow C_4 = -\frac{C_0}{12}$

at $k=3$,

$C_5 \cdot 20 + C_1 + C_1 = 0$

$\Rightarrow \quad C_5 = -\frac{C_1}{10}$

Coefficient of t^k.

$$-\left(C_{k-2}(k-2) + C_{k-2}\right) = C_{k+2}(k+2)(k+1)$$

$$\Rightarrow \quad C_{k+2} = \frac{-C_{k-2}(k-1)}{(k+2)(k+1)}, \quad (k \geq 3)$$

which is a recurrence between C_{k+2} & C_{k-2},

So the even solution,

$$u_1(t) = C_0 + C_4 t^4 + C_8 t^8 + \cdots$$

$$= C_0 - \frac{C_0}{12} t^4 + \frac{5C_0}{672} t^8 - \cdots$$

Take $C_0 = 1$, in particular

$$u_1(t) = 1 - \frac{1}{12}t^4 + \frac{5}{672}t^8 - \cdots$$

For the odd solution,

$u_2(t) = c_1 t + c_5 t^5 + c_9 t^9 + \cdots$

$\Rightarrow u_2(t) = c_1 t - \frac{c_1}{10} t^5 + \frac{c_1}{120} t^9 - \cdots$

Take, $c_1 = 1$, (particularly), we get

$u_2(t) = t - \frac{t^5}{10} + \frac{t^9}{120} - \cdots$ is the odd solution.

Now to check whether u_1 & u_2 are linearly independent

$u_1(0) = 1$, $u_1'(0) = 0$; $u_2(0) = 0$; $u_2'(0) = 1$.

$$W(0) = \begin{vmatrix} u_1(0) & u_2(0) \\ u_1'(0) & u_2'(0) \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \neq 0.$$

$\therefore$ u_1 & u_2 are linearly independent $\forall t \in \mathbb{R}$.

($\because$ Wronskian can't vanish everywhere if it's non-zero at one point).

By ratio test, for $k = n \in \mathbb{N}$

$$R = \lim_{n \to \infty} \left| \frac{C_{n-2}}{C_{n+2}} \right| = \lim_{n \to \infty} \left| \frac{(n+2)(n+1)}{(n-1)} \right| \to \infty \quad \forall t \in \mathbb{R}$$

Clearly u_1 & u_2 are converges $\forall t \in \mathbb{R}$.

1.2: Let $u(t) = \sum_{k=0}^{\infty} c_k t^k$ be the solution of the ODE:

$(1+t^2) u''(t) + u(t) = 0$, $u(0) = 1$; $u'(0) = 1$.

Now, $u'(t) = \sum_{k=0}^{\infty} c_{k+1} (k+1) t^k$

and, $(1+t^2) u''(t) = \sum_{k=0}^{\infty} c_{k+2} (k+2)(k+1) t^k (1+t^2)$

$$= \sum_{k=0}^{\infty} c_{k+2} (k+2)(k+1) t^k + \sum_{k=2}^{\infty} c_k (k)(k-1) t^k$$

Now, equating coefficient of t^k, we get

$$(k+2)(k+1)\cdot c_{k+2} + k(k-1)c_k + c_k = 0$$

$$\Rightarrow c_{k+2} = -\frac{(k(k-1)+1)}{(k+2)(k+1)} c_k \quad (k \geq 2) \quad -(i)$$

Using $u(0) = 1$, we get $c_0 = 1$

and $u'(0) = 1$, we get $c_1 = 1$

finding other coefficients,

$$\Rightarrow c_2 = \frac{-1}{2} ; \; c_3 = \frac{-1}{6} ; \; c_4 = \frac{1}{8}, \ldots$$

Therefore, unique solution satisfying $u(t) = \sum_{k=0}^{\infty} c_k t^k$ with c_k of eq(i), we get

$$u(t) = 1 + t - \frac{t^2}{2} - \frac{t^3}{6} + \frac{t^4}{8} + \cdots \quad (\text{upto } t^4)$$

$\because$ the coefficient of the ODE are analytic everywhere, so, does the solution $u(t)$ is analytic.

1.3 (a) Let $u(t) = \sum_{k=0}^{\infty} c_k t^k$ be the solution of the ODE.

Now,

$$t\,u'(t) = \sum_{k=0}^{\infty} c_{k+1}(k+1)t^{k+1} = \sum_{k=1}^{\infty} c_k \cdot k\, t^k$$

and, $(1-t^2)u''(t) = (1-t^2)\sum_{k=0}^{\infty} c_{k+2}(k+2)(k+1)t^k$

$$= \sum_{k=0}^{\infty} c_{k+2}(k+2)(k+1)t^k - \sum_{k=2}^{\infty} c_k (k)(k-1)t^k$$

Now combine terms for t^k $(k \geq 0)$

$$(k+2)(k+1)c_{k+2} - k(k-1)c_k - kc_k + \alpha^2 c_k = 0$$

$$= (k^2-\alpha^2)c_k = (k+2)(k+1)c_{k+2}$$

$$\Rightarrow \quad c_{k+2} = \frac{(k^2-\alpha^2)\ c_k}{(k+2)(k+1)} \quad , k \geq 0$$

So, there are two independent solutions,

Even solution, $\quad u_1(t) = \sum_{k=0}^{\infty} c_{2k} t^{2k}$; $c_{2(k+1)} = \frac{((2k)^2-\alpha^2)c_{2k}}{2(k+1)(2k+1)}$

and, $\quad c_{2k} = c_0 \prod_{j=0}^{k-1} \frac{((2j)^2-\alpha^2)}{(2j+2)(2j+1)}$ (Take $c_0 = 1$) in particular

and odd solution,

$$u_2(t) = \sum_{k=0}^{\infty} c_{2k+1} t^{2k+1} \quad ; \quad c_{2(k+1)+1} = \frac{(2k+1)^2-\alpha^2}{(2k+3)(2k+2)} \cdot c_{2k+1}$$

$(2j+1)$

and, $\quad c_{2k+1} = c_1 \prod_{j=0}^{j-1} \frac{(2j+1)^2-\alpha^2}{(2j+3)(2j+2)}$

(Take $c_1 = 1$) in particular.

Since, singularity of the solution
$1-t^2=0 \Rightarrow \boxed{t = \pm 1}$

So, the series u_1 & u_2 have radius of convergence ρ about a point $t=0$ which is determined by the distance from to to the nearest singularity of the ODE coefficient.

Thus, u_1 & u_2 are two linearly independent ODE in $(-1,1)$.

(b) From the reccurence relation;

$$c_{k+2} = \frac{k^2-\alpha^2}{(k+2)(k+1)} c_k \quad , \quad k \geq 0$$

Suppose $\alpha = n$ for every non-negative integer then,

$$c_{k+2} = \frac{k^2-n^2}{(k+2)(k+1)} c_k .$$

When $k = n$, $c_{n+2} = \frac{n^2-n^2}{(n+2)(n+1)} \cdot c_n = 0$

Similarly, $c_{n+4} = c_{n+6} = \cdots = 0$

i.e. the next coefficient after $c_n = 0$

Hence the power series of $u(t)$ terminates after t^n.

Thus, $u(t) = \sum_{k=0}^{n} c_k t^k$, where, $c_{k+2} = \frac{k^2-n^2}{(k+2)(k+1)} c_k$

is a polynomial solution of degree n.

1.4 (a) Let $u(t) = \sum_{k=0}^{\infty} c_k t^k$ be a solution for the ODE $u''(t) - 2tu'(t) + 2\alpha u(t) = 0$, where α is constant.

$u'(t) = \sum_{k=0}^{\infty} c_{k+1} (k+1) t^k \Rightarrow 2tu'(t) = 2\sum_{k=1} c_k \cdot k \cdot t^k$

$u''(t) = \sum_{k=0}^{\infty} c_{k+2} (k+2)(k+1) t^k$

Then substituting back to the given ODE, we get

$$\sum_{k=0}^{\infty} c_{k+2} (k+2)(k+1) t^k - 2\sum_{k=1}^{\infty} c_k \cdot k t^k + 2\alpha \sum_{k=0}^{\infty} c_k t^k = 0$$

To collect the coefficients of t^0, t^1, t^2, t^3, we get

at $k=0$, $2c_2 + 2\alpha c_0 = 0 \Rightarrow c_2 = -\alpha c_0$

at $k=1$, $6c_3 - 2c_1 + 2\alpha c_1 = 0 \Rightarrow c_3 = \frac{(1-\alpha)c_1}{3}$

at $k=2$, $12c_4 - 4c_2 + 4\alpha c_2 = 0 \Rightarrow c_4 = \frac{1}{3}(1-\alpha)c_2 = \frac{\alpha(\alpha-1)}{3} c_0$

at $k=3$, $20c_5 - 6c_3 + 2\alpha c_3 = 0 \Rightarrow c_5 = \frac{(3-\alpha)c_3}{10} = \frac{(\alpha-1)(\alpha-3)}{30} c_1$

Now, coefficient of t^k,

$$= c_{k+2}(k+2)(k+1) + 2(\alpha-k)c_k = 0$$

$$= \boxed{c_{k+2} = \frac{2(k-\alpha)\, c_k}{(k+2)(k+1)}}$$ which is a recurrence

between c_{k+2} and c_k,

So the two independent solution in $\mathbb{R}$,

Even solution, $u_1(t) = \sum_{k=0}^{\infty} c_{2k} t^{2k}$ with $c_{2(k+1)} = \frac{2(2k-\alpha)c_{2k}}{(2k+2)(2k+1)}$

and $c_{2k} = c_0 \prod_{j=0}^{k-1} \frac{2(2j-\alpha)}{(2j+2)(2j+1)}$

(Take $c_0 = 1$)

and odd solution,

$$u_2(t) = \sum_{k=0}^{\infty} c_{2k+1} t^{2k+1}, \text{ with}$$

$$c_{2(k+1)+1} = \frac{2(2k+1-\alpha)c_{2k+1}}{(2k+3)(2k+2)}$$

and, $c_{2k+1} = c_1 \prod_{j=0}^{k-1} \frac{2(2j+1-\alpha)}{(2j+3)(2j+2)}$ (Take $c_1 = 1$)

(b) From the recursive relation:

$$C_{k+2} = \frac{2(k-\alpha)C_k}{(k+2)(k+1)}$$

Suppose $\alpha = n$ for every non-negative integer,

When $k=n$, $\quad C_{n+2} = \frac{2(n-n)}{(n+2)(n+1)} \cdot C_n = 0.$

Similarly, $C_{n+4} = C_{n+6} = \dots = 0$, explicitly each term after t^n has coefficient zero.

Hence, the power series $u(t)$ terminates after t^n.

Thus, $u(t) = \sum_{k=0}^{n} C_k t^k$, where, $C_{k+2} = \frac{2(k-n)C_k}{(k+2)(k+1)}$

is a polynomial of degree n.

(c) Put $\varphi(t) = e^{-t^2}$. Then,

$$H_n(t) = (-1)^n e^{t^2} \varphi^{(n)}(t)$$

Differentiation both sides, we get

$$H_n'(t) = (-1)^n \left[2te^{t^2}\varphi^{(n)}(t) + e^{t^2}\varphi^{(n+1)}(t)\right] = 2tH_n(t) + (-1)^n e^{t^2}\varphi^{(n+1)}(t)$$

multiply last term by (-1) and use the definition of $H_{n+1}(t)$ we get

$$\Rightarrow \quad H_n'(t) = 2tH_n(t) - H_{n+1}(t) \quad (n \geq 0)$$

Now, start from $\varphi(t) = e^{-t^2}$,

as $\varphi'(t) = -2t\varphi$ and $\varphi''(t) = -2\varphi - 2t\varphi' = (4t^2-2)\varphi$

$\therefore$ Clearly $\varphi(t)$ satisfies

$$\varphi''(t) + 2t\varphi'(t) + 2\varphi(t) = 0$$

Differentiation upto n times, we get $\left(\begin{array}{l}\text{since } D^n(2t\varphi') \\ = 2t\varphi^{(n+1)} + 2n\varphi^{(n)}\end{array}\right)$

$$\varphi^{(n+2)}(t) + 2t\varphi^{(n+1)}(t) + 2(n+1)\varphi^{(n)}(t) = 0 \quad \text{---(i)}$$

<u>To show:</u> H_n satisfies the ODE $(\alpha = n)$ we need to show

$$L[H_n] := H_n'' - 2tH_n' + 2nH_n = 0$$

$$LHS = H_n'' - 2tH_n' + 2nH_n = (-1)^n e^{t^2}\left(\varphi^{n+2} + 2t\varphi^{(n+1)} + 2(n+1)\varphi^{(n)}\right)$$

$$= 0 \qquad (\text{using eq(i)})$$

We conclude that for every non-negative integer n,

$H_n(t) = (-1)^n e^{t^2} \dfrac{d^n}{dt^n}(e^{-t^2})$ is a polynomial of degree n and it satisfies the ODE with

$\alpha = n$; $\quad u'' - 2tu' + 2nu = 0$

$\square$

<u>1.5:</u> We have, $t^2 u''(t) + 2tu'(t) - 6u(t) = 0 \quad , t > 0$

Clearly this is an Cauchy-Euler homogeneous equation

Assume, $u = t^r$ be the solution. Then,

$= t^2 . r(r-1)t^{r-2} + 2t\, r\, t^{r-1} - 6t^r = 0$

$\Rightarrow r(r-1) + 2r - 6 = 0 \quad (\because t^r \neq 0)$

$= r^2 + r - 6 = 0$

which has roots $r_1 = -3$, $r_2 = 2$, which are distinct,

So, the general solution is of the form,

$$u(t) = c_1 t^{r_1} + c_2 t^{r_2} = c_1 t^{-3} + c_2 t^2, \quad (t > 0)$$

1.8: (a) Reducing the equation to standard form by $u''(t)+p(t)u'(t)+q(t)u(t)=0$ by dividing $2t(1+t)$, we get

$$p(t)=\frac{3+t}{2t(1+t)} \; ; \; q(t)=\frac{-1}{2(t+1)}$$

Clearly at $t=0$ $p(t)$ blows up so, $t=0$ is singular, & same for $t=-1$.

At $t=0$, $t\,p(t)=\frac{3+t}{1+t}$ is analytic at 0;

$$t^2 q(t)=\frac{-t^2}{2(1+t)}$$ is analytic at 0.

So, $t=0$ is an regular singular point.

Similarly, at $t=-1$ $(t+1)p(t)=\frac{3+t}{2t}$ is analytic at $t=-1$, and $(t+1)^2 q(t)=\frac{-(t+1)}{2}$ which is also analytic at $t=-1$.

So, $t=-1$, is an regular singular point.

(b) Using Frobenius Anasataz, the indicial equation is always

$$r(r-1)+P_{-1}r+q_{-2}=0, \text{ where}$$

$$P_{-1}=\lim_{t\to t_0}(t-t_0)p(t) \;\&\; q_{-2}=\lim_{t\to t_0}(t-t_0)^2 q(t)$$

where t_0 is the regular singular point.

at $t_0=0$, $P_{-1}=\lim_{t\to 0} t\cdot\frac{(3+t)}{2\cancel{t}(1+t)}=\frac{3}{2}$; $q_{-2}=\lim_{t\to 0}\frac{-t^2}{2(1+t)}=0$

So, $r(r-1)+\frac{3}{2}r=0 \Rightarrow r=0,\ r=-\frac{1}{2}$. (roots) at $t_0 = -1$,

$$P_{-1} = \lim_{t \to -1} \frac{3+t}{2t} = -1 \quad \& \quad q_{-2} = \lim_{t \to -1} -\frac{(t+1)}{2} = 0$$

So, $r(r-1) - r = 0 \Rightarrow r = 0, r = 2$ (roots)

So, the indicial equation at $t=0$ is $2r^2 + r = 0$.

and, at $t=-1$ is $r^2 - 2r = 0$

(c) Assume, $u(t) = t^r \sum_{k=0}^{\infty} a_k t^k$. (Frobenius solution at $t=0$)

After substitution into ODE the reccurrence relation is:

$$a_{n+1}(n+1+r)(2n+2r+3) + a_n(n+r)(2n+2r-1) - a_{n-1} = 0$$

<u>Case I:</u> $r = 0$

$$a_{n+1}(n+1)(2n+3) + a_n \cdot n(2n-1) - a_{n-1} = 0$$

Take $a_{-1} = 0$ and $a_0 = 1$ (By convention)

At $n=0$, $a_1(1).(3) + a_0 \cdot 0 - a_{-1} = 0 \Rightarrow a_1 = 0$

at $n=1$, $a_2(2)(5) + a_1 - a_0 = 0 \Rightarrow a_2 = \frac{1}{10}$

at $n=2$, $a_3(3)(7) + a_2 \cdot 2(3) - a_1 = 0 \Rightarrow a_3 = -\frac{1}{35}$

So, $\underset{(r=0)}{u_1(t)} = 1 + \frac{t^2}{10} - \frac{t^3}{35} + \cdots$

<u>Case II:</u> $r = -1/2$

$$a_{n+1}\left(n+\tfrac{1}{2}\right)(2n+2) + a_n\left(n-\tfrac{1}{2}\right)(2n-2) - a_{n-1} = 0$$

Take, $a_0 = 1$; at $n=0$, $a_1 = -1$; at $n=1$, $a_2 = 1/5$, ...

$\therefore$ difference of roots is not an integer.
for $(t=0)$

$$\therefore \underset{(r=-1/2)}{u_2(t)} = 1 - t + \frac{t^2}{5} - \cdots$$

So, $u_1(t)$ & $u_2(t)$ are two linearly independent solutions of the ODE around $t=0$ i.e. for $0 < |t| < R$ for some $R > 0$.

(d) Frobenius solution around $t=-1$.
$\because$ roots for the indicial polynomial around $t=-1$, is 0 and 2 with difference $= 2 \in \mathbb{Z}$
So, only the larger root $r=2$ gives a pure series and the other solution involves logarithm.

<u>Case I:</u> $r=2$,
Assume, $u(t) = (t+1)^2 \left(\sum_{k=0}^{\infty} a_k (t+1)^k \right)$

again getting a recursive relation for a_k, we get, (Take $a_0 = 1$) then, $a_1 = \frac{7}{6}$, $a_2 = \frac{53}{48}$, ...

$$\therefore u(t) = (t+1)^2 \left[1 + \frac{7}{6}(t+1) + \frac{53}{48}(t+1)^2 + \cdots \right]$$

<u>Case II</u> $r=0$.
The solution has no non-trivial solution.
So no other linear independent solution can be found around $t=-1$.